

Predicting Shipment Delays

Summary of Analysis & Strategic Recommendations

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Tasks



Analyse historical data and estimate on-time delivery



Predict the likelihood of shipment delays using historical data

Exploratory Analysis



GPS Data Challenges in Accuracy and Consistency

- **Some carriers have opted out of tracking.**
- **Data can be incomplete, with tracking stopping unexpectedly during some shipments.**
- **Carriers sometimes move closer to the destination, then away, possibly forgetting to stop tracking, i.e. the last GPS record may not be the closest to the destination.**
- **Significant variation in GPS signal frequency:**
 - **25th percentile: 0.89 signals per 5 minutes**
 - **Median: 2.42 signals per 5 minutes**
 - **75th percentile: 3.50 signals per 5 minutes.**

Exploratory Analysis



Shipment Bookings: Data Quality Challenges in Delivery and Collection Schedules & Location Data

- **Delivery and Collection times**
 - In some cases, the delivery schedule window appears to be earlier than the collection window.
 - In some cases, the latest time in a window (collection or delivery) is before the earliest time.
- **Postcodes**
 - Some entries in the shipment bookings are missing postcodes of collection or delivery.
 - External API (<https://api.opencagedata.co>) was used in order to fill the missing values by using the GPS coordinate.

First Task: Percentage of On-Time Deliveries

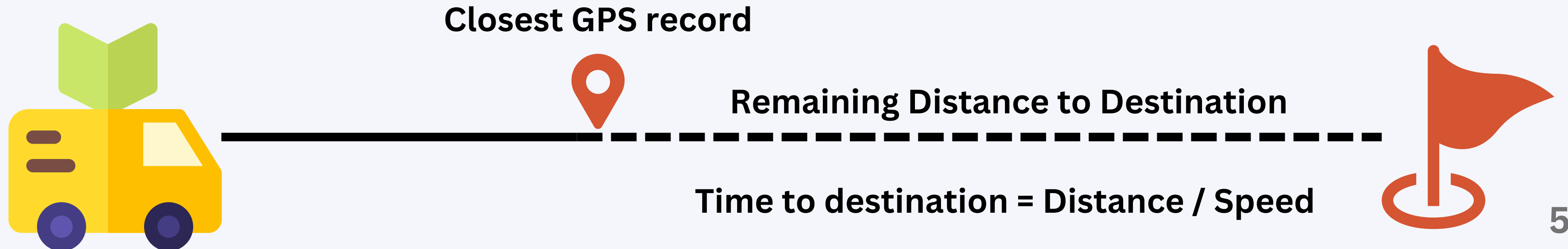
Assumptions

- Shipments with one GPS record indicates opted-out tracking.
- GPS tracking may continue after delivery, and may stop unexpectedly.
- Successful delivery occurs within 0.5 miles of the destination.
- Delivery time estimates improve as distance decreases & GPS frequency increase:
 - Certain: ≤ 0.5 miles from destination
 - High Confidence: 0.5 to 50 miles away & GPS frequency $\geq$ Median
 - Moderate Confidence: 0.5 to 50 miles away & GPS frequency $<$ Median
 - Moderate Confidence: > 50 miles away & GPS frequency $\geq$ Median
 - Low Confidence: > 50 miles away & GPS frequency $<$ Median

First Task: Percentage of On-Time Deliveries

Approach

- Identify the closest GPS record to the destination for each shipment.
- Calculate the movement speed at that GPS record and the distance remaining.
- Estimate the time needed to reach the destination using speed and distance.
- Check if the estimated arrival time fits within the delivery window.



First Task: Percentage of On-Time Deliveries

Results

- Between October 1st and December 31st, 2023
- Total shipments analysed: 3,237
- On-time arrival estimate: 68.89% (2,230 shipments)
 - 1,648 certain, 177 high confidence, 249 moderate confidence, 156 low confidence
- Late arrival estimate: 26.32% (852 shipments)
- Percentage opting out of GPS tracking: 4.79% (155 shipments)

Second Task: Likelihood of delay in new data

Pre-processing

- **Implement data pre-processing based on the insights from the previous task:**
 - **Rectify timing discrepancies in collection and delivery windows**
 - **Complete any missing postcode entries**
 - **Verify that latitude and longitude values are within a sensible range for shipments in the UK and Ireland**

Second Task: Likelihood of delay in new data

Features

Identify and extract key features that are thought to influence the prediction of whether a shipment will arrive on time or not.

1. Location-related features:

- a. The total distance between the starting and ending locations**
- b. The outer segments of the collection and delivery postcodes (to represent the area)**

2. Time-related features:

- a. The day of the week and the hour of the collection and delivery time frames**
- b. The duration of the collection and delivery time windows**

3. Additional features: Carrier ID, Vehicle Size, Vehicle Composition

Second Task: Likelihood of delay in new data

Training dataset

- **Processed historical data (Shipment_bookings.csv) to extract features for model training.**
- **Used on-time delivery estimation as ground truth (True = on time, False = delayed).**
- **Addressed data imbalance (2.6x more on-time samples) by applying oversampling to improve negative sample representation.**

Second Task: Likelihood of delay in new data

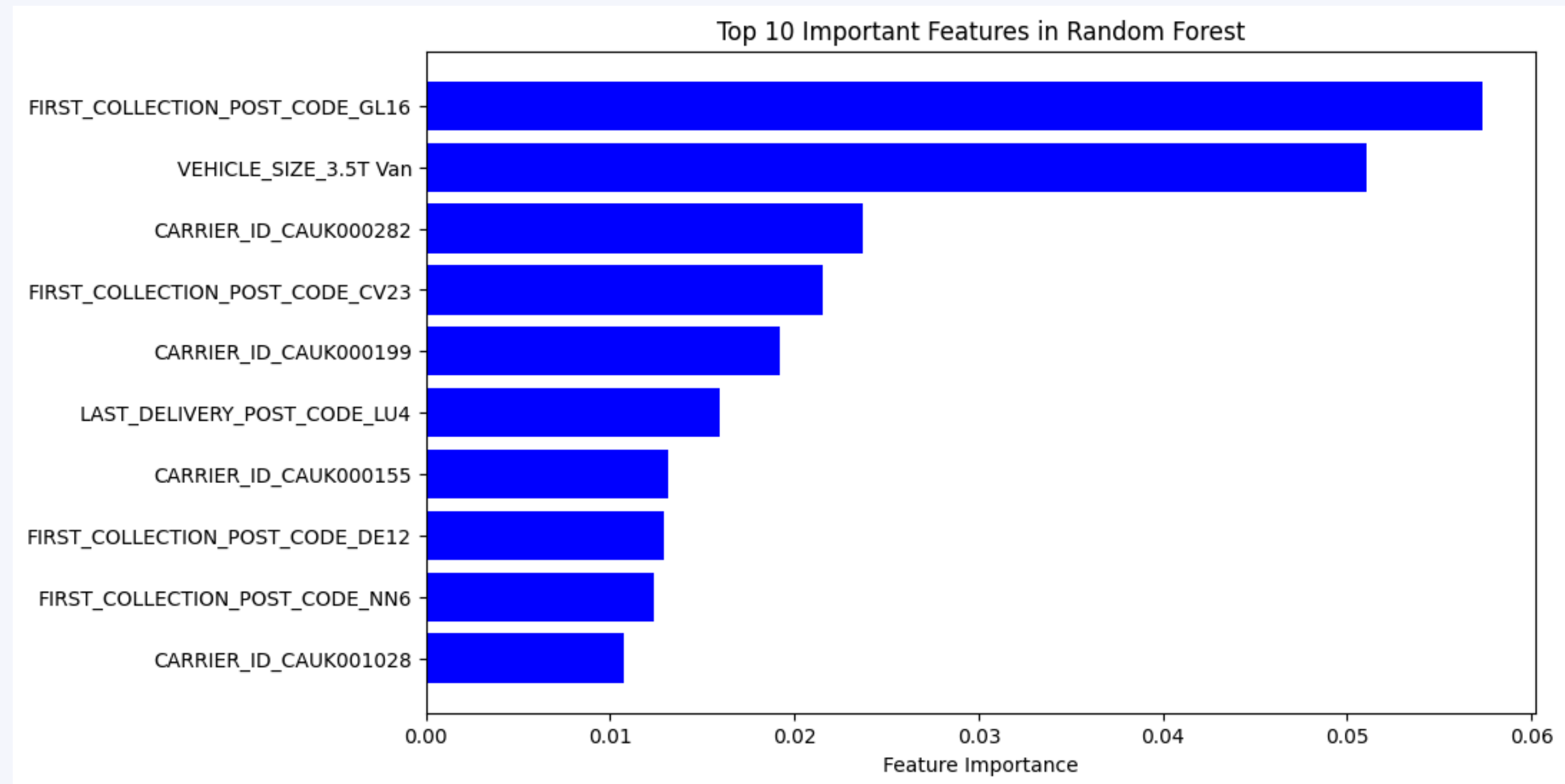
Training an AI Model to Predict Delay Likelihood

- Three predictive machine learning models were developed and evaluated:
 - **Logistic Regression:** A baseline model for binary classification, achieving an accuracy of 67.75%
 - **Random Forest:** A model that utilizes multiple decision trees, with an accuracy of 70.66%
 - **XGBoost:** An advanced model featuring additive boosting algorithm, reaching an accuracy of 71.80%
- Our conclusion is that XGBoost is the most effective model to implement.

Second Task: Likelihood of delay in new data

Features Importance

Top features contributing to shipment delay predictions include location data (postcodes), specific carriers, and vehicle types.



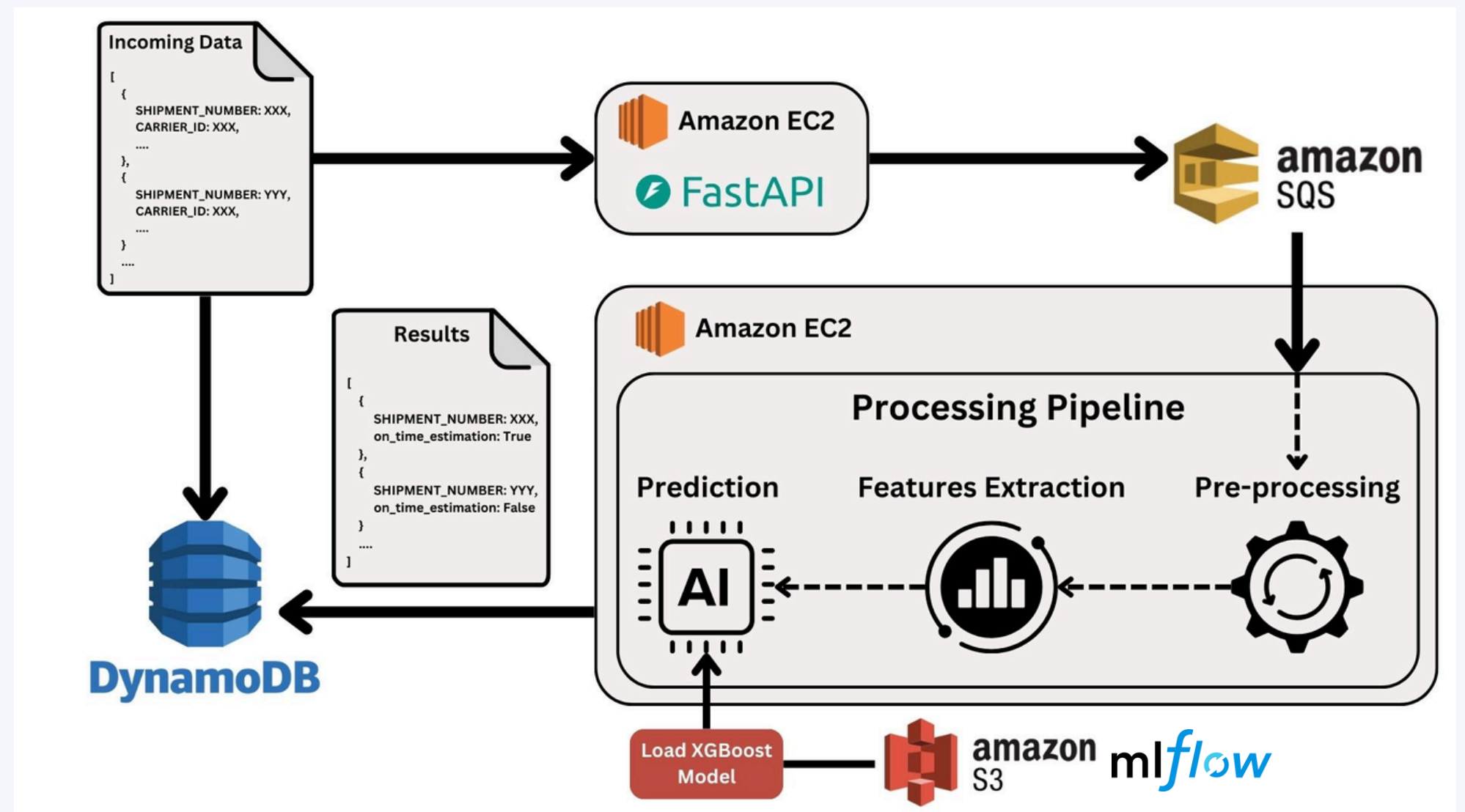
Second Task: Likelihood of delay in new data

Testing

- Pre-process and extract features from New_bookings.csv for testing.
- Use the XGBoost model to predict on-time delivery probability for the testing dataset.
- Calculate delay likelihood as $1 - \text{probability of on-time delivery}$.
- Results:
 - Shipments likely to be delayed: 265
 - Shipments likely to arrive on time: 735
 - Delay likelihood in new data: 26.5%

Technical design for deploying the prediction model through an online endpoint

- AWS Cloud
- API processes shipment requests
- Requests added to a queue
- Processing pipeline reads from the queue and performs pre-processing, feature extraction, and prediction
- Results stored in a database
- Aspects to include: Scalability, Availability, Monitoring, Logging & Security



Next Steps

Use of External Data Sources

- **Incorporate Weather Data:** Use a weather API (e.g., OpenWeather) to factor in weather conditions that could impact shipment times, enhancing model accuracy.
- **Leverage Traffic Data:** Integrate a traffic API (e.g., Google Maps Traffic) for real-time insights, improving prediction estimates by considering road conditions and congestion.
- **Improve Distance Calculations:** Utilise a Maps API (e.g., Google Maps) for precise distance calculations, rather than relying on GPS point distances, for more accurate travel estimates.

Next Steps

Data Validation

- **Introduce validation checks on shipper-entered data to ensure accuracy before submission. Automatically prompt shippers to correct any detected errors, enhancing data reliability and system efficiency.**
 - **Ensure collection and delivery time windows are logically ordered (e.g., earliest collection is before latest).**
 - **Verify the delivery window follows the collection window.**
- **Improve real-time tracking: notify if the shipper nears the destination but then moves away, asking if delivery was completed.**

Thank you

Contact

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